

D-10TM**L20017702**

Hemoglobin Testing System

Technical Bulletin: LIS Interface Requirements

D-10™ Hemoglobin Testing System

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1. INTRODUCTION

1.1. Purpose

This document defines the requirements for the communications interface between the Bio-Rad D-10 instrument and all Laboratory Information Systems (LIS).

1.2. Scope

This document is provided to the LIS vendor to aid in the design, maintenance and troubleshooting of a compatible interface between the Computer System and the Instrument. It is intended that all application-specific requirements of the interface be covered by this document. Selected generic requirements, that may be found within the LIS1-A and LIS2-A documents, are also covered.

The implementation follows LIS1-A or LIS2-A specifications.

1.3. References and Related Documents

- a) LIS1-A (formerly ASTM E1381-02) Standard Specification for Low Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems. Maintained by NCCLS.
- b) LIS2-A (formerly ASTM E1394-97) Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems. Maintained by NCCLS.

1.4. Definitions and Abbreviations

- a) Bidirectional interface - Either the Instrument or the Computer System may be the Master. The Master may initiate a message sequence that eventually causes test results to be sent to the Computer System.
- b) Instrument - Same as the D-10.
- c) Host Computer System - Same as the LIS.
- d) 1..n - Means one to many.

2. PHYSICAL LAYER

This lowest communications layer defines the physical means by which the Instrument communicates with the Computer System.

2.1. Mechanical Interface

The Instrument provides a standard DB9 male connector mounted on the back panel. The Instrument RS232 interface is configured as a DTE (Data Terminal Equipment, e.g., a terminal or PC). A DTE normally connects to a DCE (Data Communications Equipment, e.g., a modem). In this case, the Computer System is also configured as a DTE, thus a null modem cable or connector is required to hook up to the Computer System. The cable or connector may convert the 9 pin output to 25 pin male or female, as required, at the Computer System.

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Pins on the DB9 connector:

Pin Number	Name	Description
1	DCD	Data Carrier Detect. Not used.
2	RD	Receive Data. An input.
3	TD	Transmit Data. An output.
4	DTR	Data Terminal Ready. An output.
5	SGND	Ground
6	DSR	Data Set Ready. An input. Not used.
7	RTS	Request to Send. An output.
8	CTS	Clear to Send. An input.
9	RI	Ring Indicator. An input. Not used.

3. LIS1-A LAYER

This point-to-point data communications layer fits just above the Physical Layer and defines the format and content of the message packets sent back and forth from the Computer System and Instrument.

3.1. Configuration Items

The following on-site and user available configuration items control the speed and method of data transfer. Bold values are the defaults.

Baud rate	1200, 2400, 4800, 9600 baud
Data length	7 bit, 8 bit
Stop bit	1 bit , 2 bit
Parity	None , Even, Odd

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3.2. Control Characters

LIS1-A senders and receivers communicate via seven bit ASCII codes. Correct message transmission depends upon message text never including any of the below control characters.

STX	First character of any message frame.
ETX	Character immediately after the message text of a message frame. Signifies the last message frame of a group.
ETB	Character immediately after the message text of a message frame. Signifies that at least one more frame of a group is to be transmitted.
CR	Next to last character of any message frame.
LF	Last character of any message frame.
ACK	Receiver replies with ACK when it has successfully (no checksum error) received a message frame.
NAK	Receiver replies with NAK when it has unsuccessfully (checksum error or timeout) received a message frame.
ENQ	Sender sends to receiver to indicate ready to send message frame(s).
EOT	Sent by receiver to indicate that the last frame sent to receiver was received successfully, but also is a request to the sender to temporarily stop sending.

4. LIS2-A LAYER

This point-to-point data communications layer fits just above the LIS1-A Layer. The LIS2-A layer defines the format and content of the messages that may be sent back and forth from the Computer System and the Instrument.

This layer enables a bidirectional interface, meaning that either the Instrument or the Computer System may initiate the sequence of messages which eventually causes test results to be sent from the Instrument to the Computer System. The system that initiates a message sequence is known as the Master, the other system the Slave. There is no handshake or configuration required to designate a Master.

4.1. Unsupported LIS2-A Specifications

The following list contains the LIS2-A specifications that will not be implemented as part of these specifications.

- Custom escape delimiters within messages will not be supported.
- Scientific, Manufacturer, and Comment records will not be supported.
- Logical Transmission Error Recovery Requirements (LIS2-A spec, section 5.2.2).

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4.2. Record Types

Each LIS2-A message is composed of a group of the following record types. Record types within a message are separated by a carriage return delimiter.

Except for the Header and Terminator record type, record types may be repeated within a message. Level numbers restrict the ordering of record types within a message. For detailed information on level numbers, refer to the LIS2-A specification.

The format of a record type may differ depending upon the specific message type it is being used in.

Each record type contains one or more fields. Fields are separated by a delimiter. Unused fields within a record type may be skipped by inserting only the delimiter character.

In general, field data within a record should not contain embedded spaces or blanks and the following characters should not be used, as they are reserved delimiter characters used by the LIS2-A protocol:

\, /, :, *, ?, ", <, >, |, ^ &

Record Type	Record ID	Level	Description
Message Header	H	0	Instrument and Computer System identification
Patient Information	P	1	Placeholder record to maintain record hierarchy within a Test Results Message. Only the record type and sequence number fields are used.
Request Information	Q	1	Request Work List Entries
Test Order	O	2	Specific test ordered by the Computer System
Result	R	3	Results of a single test
Comment	C	1-4	Not used
Manufacturer Information	M	1-4	Not used
Scientific	S	4	Not used
Terminator	L	0	Last record of a message

4.3. Fields

Delimiter Type	Delimiter	Description
Field		Separates fields within a record
Repeat	\	Separates multiple sets of the same information within a field
Component	^	Divides one field into several subfields
Escape	&	Used to embed a special character in the data

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4.4. Test Results Query Message: Instrument ←Computer System

The purpose of the Test Results Query Message is to request that the instrument send a specific set of results stored on the instrument.

The following table lists the possible ways a Computer System may query the Instrument. Any combinations used beyond those specified in the following table shall return zero results.

NOTE: The √ indicates that the field has been specified by the user.

Starting Range ID Number	Universal Test ID	Beginning Results Date and Time	Ending Results Date and Time	Query Results
√	√			Returns all A1c or A2F test results for a specified sample
ALL	√			Returns all A1c or A2F test results for all samples
√	ALL			Returns all A1c and A2F test results for a specified sample
ALL	ALL			Returns all A1c and A2F test results for all samples
√	√	√	√	Returns all A1c or A2F test results generated between the specified date range for a specified sample
ALL	√	√	√	Returns all A1c or A2F test results generated between the specified date range for all samples
√	ALL	√	√	Returns all A1c and A2F test results generated between the specified date range for a specified sample
ALL	ALL	√	√	Returns all A1c and A2F test results generated between the specified date range for all samples

The following table provides specific information regarding the fields used in a query.

Record Type & ID	Repeat	Record Format
Message Header H	1	H Delimiter SenderName Date Time<CR>
Request Information Q	1..n	Q SeqNo StRngID UnivTestID BegDate EndDate<CR>
Terminator L	1	L SeqNo TermCode<CR>

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4.4.1. Test Results Query Message - Message Header Record

Field Name	Field Symbol	Description	Req	Typical
Record Type	H	Always H.	X	H
Delimiter Definition	Delimiter	Delimiter Definition: Field delimiter, Repeat delimiter, Component delimiter, Escape delimiter.	X	\^&
Message Control ID	Unused			
Access Password	Unused			
Sender Name or ID	SenderName	There are three components: SendName^InstNo^SoftwareVersion. Instrument number will be "LIS ID" on D-10 instrument). SendName is 1 to 16 characters containing the name of the sending system. InstNo is the LIS ID configured on the instrument. Always 2 digits in the range 0-99 zero padded. SoftwareVersion is 1 to 6 characters containing the software version of the sending system.	X	D10^02^3.00 assuming sender is Instrument. Sysmex^1^5.1 if sender is Computer System.
Sender Street Address	Unused			
ReservedField	Unused			
Sender Telephone Number	Unused			
Characteristics of Sender	Unused			
Receiver ID	Unused			
Comment or Special Instructions	Unused			
Processing ID	Unused			
Version No.	Unused			
Date and Time of Message	DateTime	The date and time the message was sent in format YYYYMMDDHHMMSS	X	20030530121314

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4.4.2. Test Results Query Message - Request Information Record

Field Name	Field Symbol	Description	Req	Typical
Record Type	Q	Always Q	X	Q
Sequence Number	SeqNo	1 to many depending on the number of query records	X	1
Starting Range ID Number	StRngID	The first component is the computer system Patient ID that is unused. The second component is the computer system specimen ID or ALL. The component is not case sensitive. For example specimen ID <i>ABC123</i> and <i>abc123</i> or <i>ALL</i> and <i>all</i> are equivalent. The computer system specimen ID is the unique specimen ID specified on the tube barcode. One to 24 characters. The keyword ALL is used to indicate all computer system specimen IDs on the instrument. This field may not be blank.	X	^344400284 ^ALL
Ending Range ID Number	Unused			
Universal Test ID	UnivTestID	There are 4 components of interest. The first three are skipped. The 4th is the Local Code and contains the test selection. Test selection is not under software control from the Computer System; it can only be changed through manual intervention at the Instrument. The test selections are: 4 (A1c) 1 (A2F)	X	^^4 ^^1
Nature of Request Time Limits	Unused			
Beginning Request Results Date and Time	BegDate	The start date in format YYYYMMDD		20030530
Ending Request Results Date and Time	EndDate	The end date in format YYYYMMDD		20030530
Requesting Physician Name	Unused			
Requesting Physican Telephone Number	Unused			

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User Field No. 1	Unused
User Field No. 2	Unused
Request Information Status Codes	Unused

4.4.3. Test Results Query Message - Terminator Record

Field Name	Field Symbol	Description	Req	Typical
Record Type	L	Always L.	X	L
Sequence Number	SeqNo	Sequence number. Always 1.	X	1
Termination Code	TermCode	Termination code. Always N. N - normal termination.	X	N

4.5. Examples - Test Results Query Message

HCS – Host Computer System (LIS)

LMS – LIS Management System (Instrument)

LIS1A control characters are in bold>.

4.5.1 Query for all A1c test results for the specified sample

```
HCS-><ENQ>
LMS-><ACK>
HCS-><STX>1H|\^&|||LIS^01^5.0|||||||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>2Q|1|^12345|^ ^ ^4<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>3L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><EOT>
```

Note: returns all A1c test results for sample 12345

4.5.2 Query for all A2F results for all samples

```
HCS-><ENQ>
LMS-><ACK>
HCS-><STX>1H|\^&|||LIS^01^5.0|||||||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>2Q|1|^ALL|^ ^ ^1<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>3L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><EOT>
```

Note: returns all A2F results on the instrument

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4.5.3 Query for all A1c and A2F results generated between the specified date range for the specified sample

```
HCS-><ENQ>
LMS-><ACK>
HCS-><STX>1H|\^&|||LIS^01^5.0|||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>2Q|1|^12345|^ ^ ^ALL|20030101|20031231<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>3L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><EOT>
```

Note: returns all A1c and A2F results for sample 12345 generated between 01/01/2003 and 12/31/2003

4.5.4 Query for all results stored on the instrument

```
HCS-><ENQ>
LMS-><ACK>
HCS-><STX>1H|\^&|||LIS^01^5.0|||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>2Q|1|^ALL|^ ^ ^ALL<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><STX>3L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
LMS-><ACK>
HCS-><EOT>
```

Note: returns all test results stored on the instrument

4.6. Test Results Message: Instrument → Computer System

The Test Results Message is sent to the Host Computer System:

- a) After each injection, after each rack, or by manual submission. When sending results to the LIS, the instrument exports all results regardless of whether or not the LIS contains a matching test order. This includes results for controls and calibrators.
- b) In response to a Test Results Query message from the Host Computer System.

It should be noted that the grouping of results sent to the Computer System is implementation dependent. Instruments are free to group multiple results in a single message or individually. Therefore, upon receiving results, Computer Systems should not make any assumptions about the way the results are grouped. For example, when sending "after each rack", Computer Systems should not assume that all of the results from the rack will be grouped together in a single message.

Also, in some cases multiple Test Order records with the same Specimen ID will be returned in a single Test Results Message. For example, a query may return results that include results for samples that were run multiple times. Therefore, in cases where Specimen ID is not enough to uniquely identify a Test Order, the Instrument Specimen ID can be used; it is guaranteed to uniquely identify a Test Order, and thus a given set of test results. The Instrument Specimen ID consists of the following fields: barcode, injection number, position in rack, timestamp, and LIS ID. The LIS ID is an identifier assigned to each D-10 instrument during the instrument's LIS configuration.

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Record Type	Repeat	Record Format
Message Header H	1	HIDelimiter SenderName Date Time<CR>
Patient P	1	PISeqNo<CR>
Test Order O	1	OISeqNoISpecimenID InstSpecID UnivTestID RptType<CR>
Result R	1..n per Test Order	RISeqNo UnivTestID MeasVal Date Time<CR>
Terminator L	1	LISeqNo TermCode<CR>

4.6.1. Test Results Message - Message Header Record

Refer to the Test Results Query Message in section 4.4.1.

4.6.2. Test Results Message - Patient Record

Zero or one Patient Record will be included in a single Test Results Message. No Patient record, and subsequently no Test Order and Result records, will be returned when results were requested by a query and no results match the criteria of the query, or if an error is encountered when processing the query. The Termination Code included in the Terminator Record can be used to determine if a query contained no results or if an error was encountered when processing the query (see section 4.6.5, Test Results Message - Terminator Record).

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Field Name	Field Symbol	Description	Req	Typical
Record Type	P	Always P.	X	P
Sequence Number	SeqNo	Sequence Number. 1 to n, varies depending on the number of records.	X	1
Practice Assigned Patient ID	Unused			
Laboratory Assigned Patient ID	Unused			
Patient ID No. 3	Unused			
Patient Name	Unused			.
Mother's Maiden Name	Unused			
Birthdate	Unused			
Patient Sex	Unused			
Patient Race-Ethnic Origin	Unused			
Patient Address	Unused			
Reserved Field	Unused			
Patient Telephone Number	Unused			
Attending Physician ID	Unused			
Special Field 1	Unused			
Special Field 2	Unused			
Patient Height	Unused			
Patient's Known or Suspected Diagnosis	Unused			
Patient Active Medications	Unused			
Patient's Diet	Unused			
Practice Field No. 1	Unused			
Practice Field No. 2	Unused			
Admission and Discharge Dates	Unused			
Admission Status	Unused			
Location	Unused			
Nature of Alternative Diagnostic Code	Unused			
Alternative Diagnostic Code	Unused			
Patient Religion	Unused			
Marital Status	Unused			
Isolation Status	Unused			
Language	Unused			
Hospital Service	Unused			
Hospital Institution	Unused			
Dosage Category	Unused			

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4.6.3. Test Results Message -Test Order Record

Field Name	Field Symbol	Description	Req	Typical
Record Type	O	Always O.	X	O
Sequence Number	SeqNo	Varies depending on the number of records.	X	1
Specimen ID	Specimen ID	This is the unique specimen ID specified on the tube barcode. One to 24 characters.	X	344400284
Instrument Specimen ID	InstSpecID	Unique ID assigned by the Instrument consisting of barcode (Specimen ID), position in rack, injection#, timestamp (YYYYMMDD), LIS ID	X	A10000027-07 007-20040109-01
Universal Test ID	UnivTestID	There are 4 components of interest. The first three are skipped. The 4th is the Local Code and contains the test selection. Test selection is not under software control from the Computer System; it can only be changed through manual intervention at the Instrument. The test selections are: 4 - This test returns A1c results. 1 - This test returns Beta-thalassemia results.	X	^^^4
Priority	Unused			
Requested/Ordered Date and Time	Unused			
Specimen Collection Date and Time	Unused			
Collection End Date and Time	Unused			
Collection Volume	Unused			
Collector ID	Unused			
Action Code	Unused			
Danger Code	Unused			
Relevant Clinical Information	Unused			
Date/Time Specimen Received	Unused			
Specimen Descriptor	Unused			
Specimen Type	Unused			
Specimen Source	Unused			
Ordering Physician	Unused			

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Field Name	Field Symbol	Description	Req	Typical
Physician's Telephone Number	Unused			
User Field No. 1	Unused			
User Field No. 2	Unused			
Laboratory Field No. 1	Unused			
Laboratory Field No. 2	Unused			
Date/Time Results Report or Last Modified	Unused			
Instrument Charge to Computer System	Unused			
Instrument Section ID	Unused			
Report Types	RptType	F or Q indicating final results. Will be set to Q when results are transmitted in response to a query.	X	F Q
Reserved Field	Unused			
Location or Ward of Specimen Collection	Unused			
Nosocomial Infection Flag	Unused			
Specimen Service	Unused			
Specimen Institution	Unused			

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4.6.4. Test Results Message - Results Record

Note: One to many Results Records can be specified for a single Test Order Record.

Field Name	Field Symbol	Description	Req	Typical
Record Type	R	Always R.	X	R
Sequence Number	SeqNo	Varies depending on the number of records.	X	1
Universal Test ID	UnivTestID	There are 5 components. The first three are skipped. The last is: Manufacturers or Local Code. Has 2 components that are always paired. Any particular Instrument run generates a subset or all of the below pairs, in any order except for Total. Total area is always the last result in the Results Record set. Further conversion of these components at the Computer System is necessary to deliver final measured values and units. There are no units specified here. The first component indicates what parameter (peak) is being measured in all cases except Total. The full set or any subset of first components may be sent in any Test Results Message. Possible values are: Unknown - unidentified peak A1a - Adult glycated hemoglobin fraction. Smaller molecule than A1b. HbA1a. A1b - Adult glycated hemoglobin fraction. Larger molecule than A1a. HbA1b. A1c - Adult glycated hemoglobin fraction. Larger molecule than A1b. HbA1c. F - Fetal hemoglobin. After birth, it decreases rapidly until only trace amounts are found in normal children and adults. HbF. LA1c - Labile A1c (Schiff base, pre-A1c). P1..Pn - Unnamed Hemoglobin. Refers to the position number of the peak on the chromatogram. P1 is the peak at the lowest retention time. E - Hemoglobin variant D - Hemoglobin variant S - Hemoglobin variant C - Hemoglobin variant Variant - for A1c Test, the E and D peaks will be labeled as Variant.	X	^^^A1a^AREA ^^^P3^TIME ^^^TOTAL^AREA

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Field Name	Field Symbol	Description	Req	Typical
		A0 - Non-glycated hemoglobin. A2 - Hemoglobin variant. Returned only when the test selection is BetaThal (Beta-thalassemia). TOTAL - Total of non-glycated and glycated hemoglobin. The second component indicates a type of HPLC output and may be: AREA - Concentration in area percent. TIME - The time elapsed between sample introduction and maximum response, i.e., retention time. Units are in minutes.		
Data or Measurement Value	MeasVal	Data or measured value.		0.79
Units	Unused			
Reference Ranges	Unused			
Result Abnormal Flags	Unused			
Nature of Abnormality Testing	Unused			
Result Status	Unused			
Date of Change in Instrument Normative Values	Unused			
Operator Identification	Unused			
Date/Time Test Started	Unused			
Date/Time Test Completed	DateTime	YYYYMMDDHHMMSS	X	20030530121314
Instrument Identification	Unused			

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4.6.5. Test Results Message - Terminator Record

Field Name	Field Symbol	Description	Req	Typical
Record Type	L	Always L.	X	L
Sequence Number	SeqNo	Sequence number. Always 1.	X	1
Termination Code	TermCode	N - normal termination. F - last request for information processed. I - no information available from last query. E - unknown internal error Q - error in last request for information.	X	N

4.7. Examples - Test Results Message

HCS – Host Computer System (LIS)

LMS – LIS Management System (Instrument)

LIS1A control characters are in bold.

4.7.1. Typical Test Results Message

```

LMS-><ENQ>
HCS-><ACK>
LMS-
><STX>1H|\^&|||D10^07^3.00|||20040223090354<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>2P|1<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>3O|1|123456|123456-02-032-20031209-
07|^^^4|||F<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>4R|1|^^^Unknown^AREA|0.9|||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>5R|2|^^^Unknown^TIME|0.15|||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-
><STX>6R|3|^^^A1b^AREA|3.5|||20031209132023<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>7R|4|^^^A1b^TIME|0.28|||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-
><STX>0R|5|^^^F^AREA|0.9|||20031209132023<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-
><STX>1R|6|^^^F^TIME|0.55|||20031209132023<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>

```

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```
LMS-><STX>2R|7|^^^A1c^AREA|4.6||| ||| |||20031209132023<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>3R|8|^^^A1c^TIME|0.82||| ||| |||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>4R|9|^^^A0^AREA|88.8||| ||| |||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>5R|10|^^^A0^TIME|1.44||| ||| |||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>6R|11|^^^TOTAL^AREA|3335057||| ||| |||20031209132023<CR>
<ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>7L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><EOT>
```

Note: The message format of results returned in response to a successful query will be the same as the format above. However, the Report Type field in each Test Order Record will be set to “Q” and the Termination Code field in the Terminator Record of the last result message sent will be set to “F” to indicate to the Host Computer that it is the last message in a set of messages returned in response to a query. See sections 4.6.3 and 4.6.5 for more information.

4.7.2. Test Results Message in response to a query with no results

```
LMS-><ENQ>
HCS-><ACK>
LMS-><STX>1H|^&|||D10^02^3.0||| ||| |||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>2L|1|I<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><EOT>
```

Note: Terminator Record’s Termination Code field set to “I” indicates no results were found that match Host Computer query.

4.7.3. Test Results Message indicating instrument query processing error

```
LMS-><ENQ>
HCS-><ACK>
LMS-><STX>1H|^&|||D10^02^3.0||| ||| |||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>2L|1|E<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><EOT>
```

Note: Terminator Record’s Termination Code field set to “E” indicates that there was an error on the instrument when processing the query.

D-10™ Hemoglobin Testing System

4.7.4. Test Results Message indicating Host Computer malformed query

```
LMS-><ENQ>
HCS-><ACK>
LMS-><STX>1H|\^&|||D10^2^000001|||20030530121314<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><STX>2L|1|Q<CR><ETX><CHK1><CHK2><CR><LF>
HCS-><ACK>
LMS-><EOT>
```

Note: Terminator Record's Termination Code field set to "Q" indicates that the host computer sent a valid LIS2-A message, but a data element inside the query did not conform to specifications. For example, a Specimen ID field (barcode) greater than 24 characters.



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